

Prithiksha K S

Electronics & Communication Engineering Undergraduate | Hardware & Embedded Systems

✉ prithikshaks@gmail.com | 📞 +91 99947 77264 | 📍 Salem, Tamil Nadu

🌐 LinkedIn | 🐙 GitHub | 🌐 Portfolio | 🚀 LeetCode

Professional Summary

ECE undergraduate (CGPA 9.19) with hands-on experience building sensor-driven embedded and IoT systems for monitoring, automation and energy applications, with skills in microcontroller interfacing and Verilog-based digital design. First Runner-Up at SAP Hackfest 2026. Eager to contribute to instrumentation, control and hardware engineering for energy and process-industry projects.

Education

B.E. Electronics and Communication Engineering	2024 – 2028
PSG Institute of Technology and Applied Research, Coimbatore	CGPA: 9.19 / 10
Higher Secondary (Class XII) – SSM Lakshmi Ammal Matric Hr. Sec. School	95.5%
Secondary (Class X) – SSM Lakshmi Ammal Matric Hr. Sec. School	97%

Technical Skills

- **Hardware & Embedded:** Arduino, Embedded C, microcontroller interfacing, sensor integration (temperature, ultrasonic/level, biometric), relay & actuator control, circuit prototyping and testing
- **Digital Design & VLSI:** Verilog HDL, combinational and sequential logic design, digital circuits
- **IoT & Monitoring:** Real-time sensor data acquisition, threshold-based alerting, Blynk dashboards
- **Programming & AI:** C, C++, Python, Machine Learning (Random Forest, CatBoost, CNN/ResNet)
- **Tools:** $\text{\LaTeX}$, MS Excel, MS PowerPoint, Canva
- **Certifications:** Digital Circuits (NPTEL), AI & ML (NPTEL), Verilog HDL (Udemy), Advanced Diploma in Python (ADPP)

Projects

- **Advanced Disruption Sensing Agents** / *AI Agents, SAP BTP, Python* **First Runner-Up, SAP Hackfest 2026**
Built a multi-agent system that monitors operational and external data signals to detect supply-chain disruptions early, assess their impact and recommend mitigation actions for faster, data-driven decisions.
- **Smart Solar Water Heating & Alert System** / *IoT, Arduino, Temperature Sensors, Blynk*
Designed a monitoring and control system for solar water heaters with real-time temperature sensing, automated control logic and mobile alerts for overheating and low-temperature conditions.
- **Water Level Management System** / *Arduino, Level Sensors, Relay Control*
Developed an automated tank-level controller using sensor feedback and control logic to switch the pump, prevent overflow and dry-run conditions, and reduce water wastage.
- **Fingerprint Voting System** / *Microcontroller, Fingerprint Sensor Module, Embedded C*
Built a biometric voting prototype with fingerprint enrolment and verification to ensure one-voter-one-vote, prevent duplicate voting and secure the election flow.
- **Driving with Wind – EV Innovation** / *Energy Harvesting, EV Systems* **Finalist, iTech Ideathon 2025**
Conceptualised a wind-energy harvesting system that uses vehicle airflow as an auxiliary power source to support EV subsystems and improve overall energy efficiency.
- **Bias Mirror – Ethical AI** / *Python, Random Forest, CatBoost, Fairlearn* **Finalist, Qualcomm Edge AI 2025**
Developed an ML-based bias detection system for recruitment data with fairness metrics and evaluation.
- **NeuroVisionAI & Radiolink-X** / *Python, Deep Learning*
Deep-learning image interpretation system and an AI radiology assistant for early tumour-risk classification.

Research Publication (Under Review)

- **Class-Aligned Deep Feature Fusion for Practical Fingerprint and Finger Vein Authentication**
Co-authored a bi-modal biometric authentication framework fusing fingerprint and finger-vein features extracted using a ResNet-based CNN, achieving accurate and spoof-resistant user authentication.

Achievements

- **First Runner-Up** – SAP Hackfest 2026 (Advanced Disruption Sensing Agents)
- **Semi-Finalist** – Caterpillar Tech Challenge 2026
- **Finalist** – Qualcomm Edge AI Hackathon 2025; iTech Ideathon 2025 (PSG iTech); Code'O Clock 24-hr Hackathon (CIT)

Leadership & Activities

- **Program Manager** – FESTO (FATC) | **Joint Secretary** – Dance Club | **Editor** – Multimedia Team, Student Union